

CHAPTER 6

Basic SQL

Basic SQL

- SQL language
 - Considered one of the major reasons for the commercial success of relational databases

SQL Data Definition

- Terminology:
 - **Table**, **row**, and **column** used for relational model terms relation, tuple, and attribute
- CREATE statement
 - Main SQL command for data definition

COMPANY relational database schema (Fig. 5.7)

EMPLOYEE

Fname	Minit	Lname	<u>Ssn</u>	Bdate	Address	Sex	Salary	Super_ssn	Dno
-------	-------	-------	------------	-------	---------	-----	--------	-----------	-----

DEPARTMENT

Dname	<u>Dnumber</u>	Mgr_ssn	Mgr_start_date
-------	----------------	---------	----------------

DEPT_LOCATIONS

<u>Dnumber</u>	<u>Dlocation</u>
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PROJECT

Pname	<u>Pnumber</u>	Plocation	Dnum
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WORKS_ON

<u>Essn</u>	<u>Pno</u>	Hours
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DEPENDENT

<u>Essn</u>	<u>Dependent_name</u>	Sex	Bdate	Relationship
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Schema and Catalog Concepts in SQL (cont'd.)

```
CREATE SCHEMA  
schemaname  
[AUTHORIZATION  
ownername] ;
```

- CREATE SCHEMA
COMPANY
AUTHORIZATION
'Jsmith' ;

- System schemas

name	schema_id	principal_id
dbo	1	1
guest	2	2
INFORMATION_SCHEMA	3	4
sys	4	4
db_owner	16384	16384
db_accessadmin	16385	16385
db_securityadmin	16386	16386
db_ddladmin	16387	16387
db_backupoperator	16389	16389
db_datareader	16390	16390
db_datawriter	16391	16391
db_denydatareader	16392	16392
db_denydatawriter	16393	16393

The CREATE TABLE Command in SQL

- Specifying a new relation
 - Provide name of table
 - Specify attributes, their types and initial constraints
- Can optionally specify schema:
 - CREATE TABLE **COMPANY**.EMPLOYEE ...
 - OR**
 - CREATE TABLE EMPLOYEE ...

Attribute Data Types and Domains in SQL

- **Basic data types**

- **Numeric data types**

- Integer numbers: INTEGER, INT, and SMALLINT
 - Floating-point (real) numbers: FLOAT or REAL, and DOUBLE PRECISION

- **Character-string data types**

- Fixed length: CHAR(*n*), CHARACTER(*n*)
 - Varying length: VARCHAR(*n*), CHAR VARYING(*n*), CHARACTER VARYING(*n*)

SQL CREATE TABLE data definition statements for defining EMPLOYEE table

EMPLOYEE

Fname	Minit	Lname	<u>Ssn</u>	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

CREATE TABLE EMPLOYEE

```
( Fname          VARCHAR(15)          NOT NULL,
  Minit          CHAR,
  Lname          VARCHAR(15)          NOT NULL,
  Ssn            CHAR(9)              NOT NULL,
  Bdate          DATE,
  Address        VARCHAR(30),
  Sex            CHAR,
  Salary         DECIMAL(10,2),
  Super_ssn      CHAR(9),
  Dno            INT                  NOT NULL,
  PRIMARY KEY (Ssn),
```


SQL CREATE TABLE data definition statements for defining the COMPANY schema from Figure 5.7 (Fig. 6.1)

```
CREATE TABLE DEPARTMENT
  ( Dname          VARCHAR(15)          NOT NULL,
    Dnumber        INT                  NOT NULL,
    Mgr_ssn        CHAR(9)             NOT NULL,
    Mgr_start_date DATE,
  PRIMARY KEY (Dnumber),
  UNIQUE (Dname),
  FOREIGN KEY (Mgr_ssn) REFERENCES EMPLOYEE(Ssn) );

CREATE TABLE DEPT_LOCATIONS
  ( Dnumber        INT                  NOT NULL,
    Dlocation      VARCHAR(15)         NOT NULL,
  PRIMARY KEY (Dnumber, Dlocation),
  FOREIGN KEY (Dnumber) REFERENCES DEPARTMENT(Dnumber) );
```

continued on next slide

SQL CREATE TABLE data definition statements for defining the COMPANY schema from Figure 5.7 (Fig. 6.1)-continued

CREATE TABLE PROJECT

(Pname	VARCHAR(15)	NOT NULL,
Pnumber	INT	NOT NULL,
Plocation	VARCHAR(15),	
Dnum	INT	NOT NULL,

PRIMARY KEY (Pnumber),

UNIQUE (Pname),

FOREIGN KEY (Dnum) **REFERENCES** DEPARTMENT(Dnumber));

CREATE TABLE WORKS_ON

(Essn	CHAR(9)	NOT NULL,
Pno	INT	NOT NULL,
Hours	DECIMAL(3,1)	NOT NULL,

PRIMARY KEY (Essn, Pno),

FOREIGN KEY (Essn) **REFERENCES** EMPLOYEE(Ssn),

FOREIGN KEY (Pno) **REFERENCES** PROJECT(Pnumber));

CREATE TABLE DEPENDENT

(Essn	CHAR(9)	NOT NULL,
Dependent_name	VARCHAR(15)	NOT NULL,
Sex	CHAR,	
Bdate	DATE,	
Relationship	VARCHAR(8),	

PRIMARY KEY (Essn, Dependent_name),

FOREIGN KEY (Essn) **REFERENCES** EMPLOYEE(Ssn));

Specifying Key and Referential Integrity Constraints

- **PRIMARY KEY** clause

- Specifies one or more attributes that make up the primary key of a relation
- `Dnumber INT PRIMARY KEY;`

- **UNIQUE** clause

- Specifies alternate (secondary) keys (CANDIDATE keys).
- `Dname VARCHAR(15) UNIQUE;`

Specifying Key and Referential Integrity Constraints (cont'd.)

- **FOREIGN KEY** clause
 - Default operation: reject update on violation
 - Attach **referential triggered action** clause
 - Options include SET NULL, CASCADE, and SET DEFAULT
 - Action taken by the DBMS for SET NULL or SET DEFAULT is the same for both ON DELETE and ON UPDATE
 - CASCADE option suitable for “relationship” relations

The SELECT-FROM-WHERE

Structure of Basic SQL Queries

- Basic form of the `SELECT` statement:

```
SELECT    <attribute list>  
FROM      <table list>  
WHERE     <condition>;
```

where

- <attribute list> is a list of attribute names whose values are to be retrieved by the query.
- <table list> is a list of the relation names required to process the query.
- <condition> is a conditional (Boolean) expression that identifies the tuples to be retrieved by the query.

Basic Retrieval Queries

Query 0. Retrieve the birth date and address of the employee(s) whose name is 'John B. Smith'.

EMPLOYEE

Fname	Minit	Lname	<u>Ssn</u>	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

Basic Retrieval Queries

Query 0. Retrieve the birth date and address of the employee(s) whose name is 'John B. Smith'.

Q0: **SELECT** Bdate, Address
 FROM EMPLOYEE
 WHERE Fname='John' AND Minit='B' AND Lname='Smith';

EMPLOYEE

Fname	Minit	Lname	<u>Ssn</u>	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

Output:

<u>Bdate</u>	<u>Address</u>
1965-01-09	731 Fondren, Houston, TX

Basic Retrieval Queries

Query 1. Retrieve the name and address of all employees who work for the 'Research' department.

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

DEPARTMENT

Dname	Dnumber	Mgr_ssn	Mgr_start_date
Research	5	333445555	1988-05-22
Administration	4	987654321	1995-01-01
Headquarters	1	888665555	1981-06-19

Basic Retrieval Queries

Query 1. Retrieve the name and address of all employees who work for the 'Research' department.

Q1: **SELECT** Fname, Lname, Address
 FROM EMPLOYEE, DEPARTMENT
 WHERE Dname='Research' **AND** Dnumber=Dno;

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

DEPARTMENT

Dname	Dnumber	Mgr_ssn	Mgr_start_date
Research	5	333445555	1988-05-22
Administration	4	987654321	1995-01-01
Headquarters	1	888665555	1981-06-19

Output:

Fname	Lname	Address
John	Smith	731 Fondres, Houston, TX
Franklin	Wong	638 Voss, Houston, TX
Ramesh	Narayan	975 Fire Oak, Humble, TX
Joyce	English	5631 Rice, Houston, TX

Basic Retrieval Queries (Contd.)

Query 2. For every project located in 'Stafford', list the project number, the controlling department number, and the department manager's last name, address, and birth date.

PROJECT

Pname	<u>Pnumber</u>	Plocation	Dnum
ProductX	1	Bellaire	5
ProductY	2	Sugarland	5
ProductZ	3	Houston	5
Computerization	10	Stafford	4
Reorganization	20	Houston	1
Newbenefits	30	Stafford	4

DEPARTMENT

Dname	<u>Dnumber</u>	Mgr_ssn	Mgr_start_date
Research	5	333445555	1988-05-22
Administration	4	987654321	1995-01-01
Headquarters	1	888665555	1981-06-19

EMPLOYEE

Fname	Minit	Lname	<u>Ssn</u>	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

Basic Retrieval Queries (Contd.)

Query 2. For every project located in 'Stafford', list the project number, the controlling department number, and the department manager's last name, address, and birth date.

Q2: **SELECT** Pnumber, Dnum, Lname, Address, Bdate
 FROM PROJECT, DEPARTMENT, EMPLOYEE
 WHERE Dnum=Dnumber **AND** Mgr_ssn=Ssn **AND**
 Plocation='Stafford';

PROJECT

Pname	<u>Pnumber</u>	Plocation	Dnum
ProductX	1	Bellaire	5
ProductY	2	Sugarland	5
ProductZ	3	Houston	5
Computerization	10	Stafford	4
Reorganization	20	Houston	1
Newbenefits	30	Stafford	4

DEPARTMENT

Dname	<u>Dnumber</u>	Mgr_ssn	Mgr_start_date
Research	5	333445555	1988-05-22
Administration	4	987654321	1995-01-01
Headquarters	1	888665555	1981-06-19

EMPLOYEE

Fname	Minit	Lname	<u>Ssn</u>	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

Output:

<u>Pnumber</u>	<u>Dnum</u>	<u>Lname</u>	<u>Address</u>	<u>Bdate</u>
10	4	Wallace	291Berry, Bellaire, TX	1941-06-20
30	4	Wallace	291Berry, Bellaire, TX	1941-06-20

Aliasing, and Renaming

- **Aliases or tuple variables**

- Declare alternative relation names E and S to refer to the EMPLOYEE relation twice in a query:

Query 8. For each employee, retrieve the employee's first and last name and the first and last name of his or her immediate supervisor.

```
SELECT E.Fname, E.Lname, S.Fname, S.Lname  
FROM EMPLOYEE AS E, EMPLOYEE AS S  
WHERE E.Super_ssn=S.Ssn;
```

Substring Pattern Matching and Arithmetic Operators

- **LIKE** comparison operator
 - Used for string **pattern matching**
 - % replaces an arbitrary number of zero or more characters
 - underscore (_) replaces a single character
 - Examples: **WHERE** Address **LIKE** '%Houston,TX%';
 - **WHERE** Ssn **LIKE** '__ 1__ 8901';
- **BETWEEN** comparison operator

E.g., in Q14 :

WHERE(Salary **BETWEEN** 30000 **AND** 40000)
AND Dno = 5;

Basic Retrieval Queries

- Query:** Show the resulting salaries if every employee working on the 'ProductX' project is given a 10 percent raise.

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

WORKS_ON

Essn	Pno	Hours
123456789	1	32.5
123456789	2	7.5
666884444	3	40.0
453453453	1	20.0
453453453	2	20.0
333445555	2	10.0
333445555	3	10.0
333445555	10	10.0
333445555	20	10.0
999887777	30	30.0
999887777	10	10.0
987987987	10	35.0
987987987	30	5.0
987654321	30	20.0
987654321	20	15.0
888665555	20	NULL

PROJECT

Pname	Pnumber	Plocation	Dnum
ProductX	1	Bellaire	5
ProductY	2	Sugarland	5
ProductZ	3	Houston	5
Computerization	10	Stafford	4
Reorganization	20	Houston	1
Newbenefits	30	Stafford	4

Basic Retrieval Queries

- *Query:* **Show the resulting salaries if every** employee working on the 'ProductX' project is given a **10 percent raise**.

```
SELECT E.Fname, E.Lname, 1.1 * E.Salary AS Increased_sal
FROM EMPLOYEE AS E, WORKS_ON AS W, PROJECT AS P
WHERE E.Ssn=W.Essn AND W.Pno=P.Pnumber AND
P.Pname='ProductX';
```

Output

Fname	Lname	Increased_sal
John	Smith	33000
Joyce	English	27500

Basic SQL Retrieval Query Block

```
SELECT    <attribute list>  
FROM      <table list>  
[ WHERE    <condition> ]  
[ ORDER BY <attribute list> ];
```


INSERT, DELETE, and UPDATE Statements in SQL

- Three commands used to modify the database:
 - INSERT, DELETE, and UPDATE
- INSERT typically inserts a tuple (row) in a relation (table)
- UPDATE may update a number of tuples (rows) in a relation (table) that satisfy the condition
- DELETE may also update a number of tuples (rows) in a relation (table) that satisfy the condition

The INSERT Command

- Specify the relation name and a list of values for the tuple. All values including nulls are supplied.

```
U1:  INSERT INTO  EMPLOYEE
      VALUES      ( 'Richard', 'K', 'Marini', '653298653', '1962-12-30', '98
                    Oak Forest, Katy, TX', 'M', 37000, '653298653', 4 );
```

- The variation below inserts multiple tuples where a new table is loaded values from the result of a query.

```
U3B:  INSERT INTO  WORKS_ON_INFO ( Emp_name, Proj_name,
                                     Hours_per_week )
      SELECT        E.Lname, P.Pname, W.Hours
      FROM          PROJECT P, WORKS_ON W, EMPLOYEE E
      WHERE         P.Pnumber=W.Pno AND W.Essn=E.Ssn;
```

The DELETE Command

- Removes tuples from a relation
 - Includes a `WHERE` clause to select the tuples to be deleted. The number of tuples deleted will vary.

U4A:	DELETE FROM	EMPLOYEE
	WHERE	Lname='Brown';
U4B:	DELETE FROM	EMPLOYEE
	WHERE	Ssn='123456789';
U4C:	DELETE FROM	EMPLOYEE
	WHERE	Dno=5;
U4D:	DELETE FROM	EMPLOYEE;

UPDATE (contd.)

- Example: **Change** the location and controlling department number of project number 10 to 'Bellaire' and 5, respectively

```
U5:      UPDATE    PROJECT
          SET       PLOCATION = 'Bellaire',
                  DNUM = 5
          WHERE     PNUMBER=10
```

UPDATE (contd.)

- Example: **Give** all employees in the 'Research' department a 10% raise in salary.

```
U6:UPDATE      EMPLOYEE
      SET       SALARY = SALARY *1.1
      WHERE     DNO IN (SELECT  DNUMBER
                           FROM    DEPARTMENT
                           WHERE   DNAME='Research')
```

- In this request, the modified SALARY value depends on the original SALARY value in each tuple
 - The reference to the SALARY attribute on the right of = refers to the old SALARY value before modification
 - The reference to the SALARY attribute on the left of = refers to the new SALARY value after modification